INCH-POUND

MIL-B-49450(ER)
AMENDMENT 3
23 July 1992
SUPERSEDING
AMENDMENT 2
19 July 1988

MILITARY SPECIFICATION
BATTERY, RECHARGEABLE, NICKEL-CADMIUM,
VENTED, AIRCRAFT, BB-716/A

This amendment forms a part of MIL-B-49450(ER), dated 16 January 1987, and is approved for use within the Laboratory Command, Department of the Army and is available for use by all Departments and Agencies of the Department of Defense.

PAGE 1

Insert after title:

"This specification is approved for use within Laboratory Command, Department of the Army, and is available for use by all Department and Agencies of the Department of Defense."

PAGE 3

* Add FED-STD-751 Stitches, Seams and Stitchings

PAGE 5

3.4.1.1, fifth line, delete ".15" and substitute "0.015".

PAGE 6

3.4.2.5, delete last sentence and substitute:

"The vent plug shall withstand 7 in-lb. of torque with no damage."

PAGE 7

3.4.2.6, last line, insert the word "potassium" before carbonate.

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3.4.3.1, delete second and third sentences and substitute:

"The thermistor used shall conform to MIL-T-23648. The thermistor shall be wired per applicable specification sheet."

3.4.3.2, delete in its entirety and substitute:

"3.4.3.2 Sensor Receptacle. The battery shall use a hermetic, jam nut receptacle, per MIL-C-26482, series 2, with inconel shell and contacts, which mates with connector MS-3476-L12-10S. The receptacle shall be provided with a special "O" ring, Marathon Battery Co. part number 10343-9 or equal, of ethylene propylene rubber with an inner diameter of 7/8 inch, an outer diameter of 1 1/8 inch and a wall thickness of 0.139 plus 0.004, -0.000 inch."

3.4.3.3, fourth line, after MIL-S-24236/1-SKKKG, insert "or Elmwood Sensors Inc. part number 3153-23-19 or equivalent."

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* 3.4.10, delete in its entirety and substitute:

"3.4.10 Carrying Strap. The battery cover shall have a permanently attached carrying strap made of web nylon per MIL-W-4088. The strap attachment to the cover shall either be by riveting per requirement 12, MIL-STD-454 or by using a strap connector which is welded to the battery cover per requirement 13, MIL-STD-454. The strap shall be attached to the strap connector by sewing IAW FED-STD-751, Type 301, 8-12 threads per inch using nylon thread per V-T-295, Type I or II, Class I. All exposed cut edges of the web strap shall be heat sealed or use equivalent methods to prevent unraveling."

PAGE 9

3.5.2.1, 6th line, delete "Maximum" and substitute "Minimum".

PAGE 10

3.5.2.4, delete ".200±.030 inch lettering" and substitute "0.17 inch lettering"; delete ".120±.030 inch lettering" and substitute "0.09 inch lettering, minimum".

PAGE 11

3.5.2.7, note 2, delete in its entirety and substitute:

"2. Do not allow caustic electrolyte to contact body, clothing, tools, etc. If contact is made wash immediately with water. Flush eyes with water only for 15 minutes and seek medical attention."

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3.5.2.8, Instruction 4, line 3, delete "plate" and substitute "plates", 2 places.

PAGE 17

4.6.1.2.1, line 4, delete "distilded" and substitute "distilled or deionized".

PAGE 19

4.7.4.2, delete first sentence and substitute:

"The vent plug of each cell being tested shall be closed to a torque of 7 in-lb and then opened."

4.7.4.2, add the following sentence:

"The vent plugs may be screwed into a threaded metal adapter, rather than a cell, to conduct this test."

PAGE 21

4.7.14, delete first sentence and substitute:

"The battery shall be charged per 4.6.1.2.1 and subjected to the thermal shock test per procedure 1, Method 503.2 of MIL-STD-810 using a low temperature of -70°F (-57°C) and a high temperature of 160°F (71°C)."

4.7.14, insert the following after the first sentence:

"Diurnal cycling per procedure 1, Method 503.2 of MIL-STD-810 is not required."

PAGE 22

4.7.16, delete in its entirety and substitute:

"4.7.16 Humidity and charge retention. The battery shall be charged per 4.6.1.2.1 and then subjected to the humidity test per Procedure II, Method 507.1 of MIL-STD-810. The battery case air vents shall be positioned where the axis of the vents are parallel to the chamber air flow. Upon completion, accumulated water may be removed from the battery. The battery shall be discharged at the rate specified by the applicable specification sheet to the specified cut-off voltage and shall meet the requirements of 3.6.1 and 3.6.2.

4.7.21, second line, delete "MS3474-L12-10PN" and substitute "MIL-C-26482, Series 2."

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Table II, Receptacle, delete "MS3474-L12-10PN" and substitute "MIL-C-26482".

Table II, Polyamide material, delete "PA811" and add "PA611".

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Table III, perform temperature shock of harness test prior to alkali resistance of harness test.

PAGE 32

* Table VI, number 16 description, add to the end: "Carrying strap stitches are loose or broken. Strap cut edges are not sealed to prevent unraveling."

Table VI, number 17, delete "MS3474L12-10PN" and substitute "MIL-C-26482, Series 2 connector."

The margins of this amendment are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous amendment were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous amendment.

Preparing Activity:
Army - ER

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